

1. Decimal 0.829

a) Expanded form using powers of 10

$$0.829 = \frac{8}{10} + \frac{2}{100} + \frac{9}{1000} = 0.8 + 0.02 + 0.009$$

b) Number line representation

0.829 lies **between 0.82 and 0.83**, closer to 0.83.

c) Finite or infinite?

Finite decimal, because it ends after three decimal places.

2. Convert fractions to decimals

1) $\frac{32}{99}$

= **0.323232...** (infinite repeating)

✗ Not finite (denominator has factor 3)

2) $\frac{125}{99}$

a) Denominator factorisation

$$99 = 3 \times 3 \times 11$$

Since it has factors other than 2 or 5 → **not finite**

b) Decimal form

$$\frac{125}{99} = 1.262626...$$

Infinite repeating

3) $\frac{15}{128}$

$$\frac{15}{128} = 0.1171875$$

✓ Finite

4) $\frac{15}{3} = 5$

✓ Finite

5) $\frac{28}{13}$

Infinite (13 ≠ 2 or 5)

6) $4005 = 80 \frac{400}{5} = 805400 = 80$

✓ Finite

7) $6415 \frac{64}{15} 1564$

Infinite ($15 = 3 \times 5$)

8) $35199 \frac{35}{199} 19935$

Infinite

9) $250219 \frac{250}{219} 219250$

Infinite

10) $6251000 = 0.625 \frac{625}{1000} = 0.6251000625 = 0.625$

✓ Finite

3. Decimal 0.55555...

a) Expanded form

$$0.55555... = 510 + 5100 + 51000 + ...$$

$$0.55555... = \frac{5}{10} + \frac{5}{100} + \frac{5}{1000} + ...$$

$$0.55555... = 105 + 1005 + 10005 + ...$$

b) Number line

Between **0.5** and **0.6**, getting closer to **0.56**

c) Finite or infinite?

Infinite repeating, because 5 repeats forever.

4. Decimal 0.573

a) Expanded form

$$0.573 = 510 + 7100 + 310000$$

$$0.573 = \frac{5}{10} + \frac{7}{100} + \frac{3}{1000}$$

$$0.573 = 105 + 1007 + 10003$$

b) Number line

Between **0.57** and **0.58**

c) Finite or infinite?

Finite, because it ends.

5. Decimal 0.625

a) Expanded form

$$0.625 = \frac{6}{10} + \frac{2}{100} + \frac{5}{1000} \quad 0.625 = 0.6 + 0.02 + 0.005$$

b) Number line

Between **0.62** and **0.63**

c) Finite or infinite?

Finite

6. Decimal 0.370370...

a) Expanded form

$$0.370370... = \frac{3}{10} + \frac{7}{100} + \frac{0}{1000} + ... \quad 0.370370... = 0.3 + 0.07 + 0.000370... \\ 0.370370... = 0.3 + 0.07 + 0.000370... + ...$$

b) Number line

Between **0.37** and **0.38**

c) Finite or infinite?

Infinite repeating

7. Which is more accurate: 0.6666 or 0.6?

- **0.6666** is more accurate (closer to $\frac{2}{3}$)
- **0.6** is easier to compute with
- Infinite decimals are shortened to finite ones for calculations

8. Why shorten infinite decimals?

- Infinite decimals cannot be fully written
- Shortening gives **approximate answers**
- More decimal places → more accuracy

9. Prove that $0.999... = 1$

- Each extra 9 adds a smaller value
- The number gets closer and closer to 1
- Therefore,

$$0.999\dots = 10.999\dots = 10.999\dots = 1$$

10. Why is **0.33333** > **0.3333**?

$$0.33333 = 0.3333 + 0.00003 \quad 0.33333 = 0.3333 + 0.00003 \quad 0.33333 = 0.3333 + 0.00003$$

So **0.33333** is larger